



Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering

Course Code & Title: CB3491 Cryptography and Cyber Security

Name of the Faculty member (s): Mrs.K. Jeyakarthika, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Classical Encryption Technique- Hill Cipher
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Collaborative Learning
- **Justification:**
 - Collaborative learning, a dynamic teaching approach, fosters student engagement by organizing them into groups to collectively tackle challenges.
 - Through this method, students pool their diverse perspectives and expertise to solve problems, enhancing their understanding of complex concepts while promoting intrinsic motivation.
 - It also helps students make concepts more interesting by involving them to work together.
- **Time Allotted for the Activity:** 40 min
- **Details of the Implementation**
 - The collaborative learning activity provide students the best opportunities to practice the tasks in an authentic way, including interactions and working with other people.
 - The students were divided into small groups, ensuring each group comprised 3 to 4 members with total 13 groups
 - The task of solving classical encryption technique – Hill cipher with key value 3×3 matrix was assigned to each team.
 - Team members started working on the assigned task and discussed with them. It is shown in Fig.1.
 - Each group was provided with an opportunity to present their solution regarding to the given task.
 - Team members explained the example with detailed solution for Hill cipher is shown in Fig.2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12
CO1	3	2	1	2	3	1	1	1

- **PO / PSO mapped:**

Innovation practice	PO1	PO2	PO3	PO4	PO5	PO8
	3	2	1	2	3	1
Justification for correlation	It involves logical thinking and make the students able to think about problem solving approach	Students will be able to understand the problem than theoretical explanation	Students will be able to provide solution for the problem by understand the problem through this activity	Student will be able to analyze the given problem	Use of virtual environment and crypt tool for performing encryption and decryption is highly correlated	Students will complete the assignments/ tasks in an ethical way
	PO9	PO12	PSO 1	PSO 2		
	1	1	2	3		
	As the students are involved in the activity as team of two, the team work skill improved	Ability to reproduce the contents gathered through self learning	The problem-solving skill earned through this activity helps the students in solving various problems	The concept is used in security applications		

- **Images/Screenshot of the practice**



Fig.1. Team Discussion



Fig.2. Presentation by Students

- **Reflective Critique:**

- **Feedback of practice from students and other stakeholders:**

- Student felt good, since they learn the concepts as a group.
- They felt that through such learning, Students actively participate, and this activity encouraged the students to share their knowledge with others.

- **Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)**

- Students are actively participated in each group.
- It helps the students are made to work in a team and share their ideas to others and improve their communication skills.
- It helps to focus attention and engage students in learning.
- This activity provided students with a deeper understanding of the specific topic.
- Since it is Problematic concept students understand well while discussing and solving among themselves.

- **Challenges faced in implementation:**

- It takes more time to make all the teams present the answers to the given questions.
- Some of the team members hesitate to present in front of the class.

References:

- <https://www.opencolleges.edu.au/informed/features/pecr-teaching/>
- <https://tilt.colostate.edu/fipsAndGuides/fip/180>
- <https://www.valamis.com/hub/collaborative-learning>
- https://www.ritrjpm.ac.in/images/computer-science/20232024/IP/4.KVS_23_24_CS3352_CL.pdf